Department of Computer Science and Engineering
Shiv Nadar University Chennai
July-November 2026 Semester
M.Tech. Course on Deep Learning
Lab Assignment 3

Date: 18th August, 2026

1. **For 36-dimensional data of 5 image classes in Dataset2, build a multilayer feedforward neural network with two hidden layers with 30 nodes in the first hidden layer and 15 nodes in the second hidden layer. Use the softmax as the activation function in the output layer and the cross-entropy error as the error function. The stopping criterion is that the change in the average training error should be below a suitable threshold.**

   Configurations:

   Configuration1: Hyperbolic tangent function with $\beta = 1.0$ as the activation function in the hidden layers

   Configuration2: ReLU function as the activation function in the hidden layers.

   Modes of learning:

   1. Pattern mode (Stochastic gradient descent)
   2. Mini-batch mode with the number of subsets as 5

   Parameter update rule:

   1. Delta rule with $\eta = 0.8$
   2. Generalized delta rule with $\eta = 0.8$ and $\alpha = 0.9$
   3. AdaM rule

   For each of the combinations of the configurations, modes of learning and parameter update rules, give the following:

   1. Error vs Epoch plots for (i) the training data error and (ii) the validation data error.

   2. Classification accuracy and confusion matrix for (i) training data, (ii) validation data and (iii) test data.

   For the combination of Configuration2, Mini-batch mode of learning and AdaM rule, train the model with each of the following normalization methods used separately, and present the above mentioned plots, accuracies and confusion matrices:

   1. Mean-variance normalization of input features
   2. Layer normalization